Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 2 Random variable and their probability distributions

Section 2.4-1 Distribution of random variable function

- Definition
- Distribution of functions of discrete random variables
- Distribution of functions of continuous random variables

■ Section 2.4-1 Distribution of random variable function

- In this section we will study a bit more complex random variables. If we know the length A of a side of a square is an r.v., and we know its distribution, the squares area A^2 is also an r.v., then we would like to know what is the distribution of A^2.

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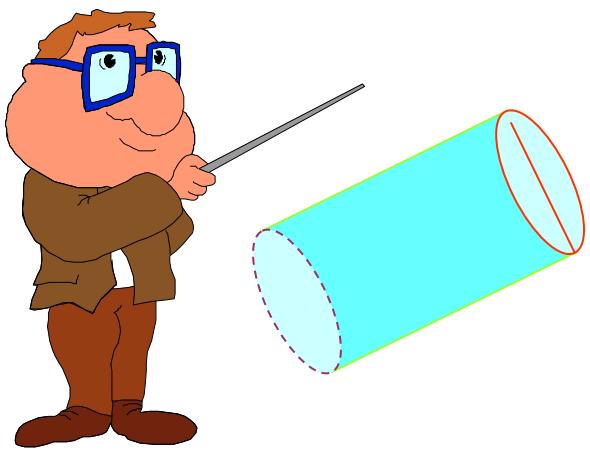
In many practical problems, it is necessary to study the functional relationship between random variables, that is, their relationship in probability distribution

For example: The distribution of the cross-sectional diameter *d* of the circular shaft is known,



Now find the distribution of the cross - sectional area:

$$A=\frac{\pi d^2}{4}$$

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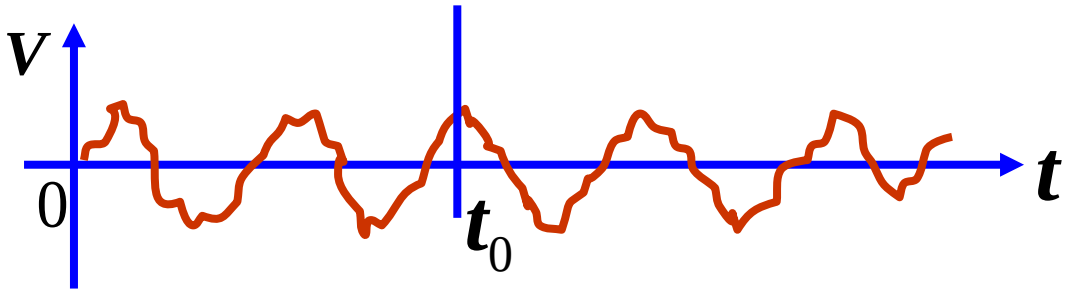
Example 2： The distribution of noise voltage V at time t = t_0 is known, find the distribution of power $W=V^2/R$ (R is resistance), etc.



■ Section 2.4-1 Distribution of random variable function

- More generally, X is an r.v. and the function $y = g(x)$ is continuous, then $Y = g(X)$ is also an r.v., and we call it the function of random variable X.

- If the distribution of X is given, what's the distribution of Y = g(X). If X is discrete, this is very simple. Another case is for continuous random variables. We note that for continuous random variables, the derivative of the cdf is just the pdf.

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Definition: Let $g(x)$ be a function defined on the set of all possible values x of random variable X. if for each possible value x of X, there is a corresponding value $y = g(x)$ of another random variable Y, Y is called the function of X, which is recorded as $Y = g(X)$.

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Task of this section: calculate the distribution of Y according to the distribution of X, or calculate the distribution of $y = f(x_1, x_2, \ldots, x_n)$ from the distribution of $(x_1, x_2, \ldots, x_n)$.

This problem is important both in practice and in theory.

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■ Section 2.4-1 Distribution of random variable function

If X is a discrete random variable, Y=g(X) is also a discrete random variable, and the distribution of g(X) can be obtained directly from the distribution of X.

Example 1

The probability distribution law of r.v *X* is known as:

X	5	10
P_k	$\frac{1}{3}$	$\frac{2}{3}$

Find: The distribution law of r.v *Y*, $Y = 2X$

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Solution: The possible values of X are: $x_1 = 5, \quad x_2 = 10$

The possible values of Y are: $y_1 = 10, \quad y_2 = 20$

and, $$P(Y=10) = P(2X=10) = P(X=5) = \frac{1}{3}$$

$$P(Y=20) = P(2X=20) = P(X=10) = \frac{2}{3}$$

Therefore, the distribution law of $Y = 2X$ is:

Y	10	20
P_k	$\frac{1}{3}$	$\frac{2}{3}$

X	5	10
P_k	$\frac{1}{3}$	$\frac{2}{3}$

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Example 2

The probability distribution of X is known as:

X	0	1	2	3	4	5
P_k	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{2}{9}$	$\frac{1}{9}$

Find: $Y=(X-2)^2$ **probability distribution.**

Solution:

Y	0	1	4	9
P_k	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{11}{36}$	$\frac{1}{9}$

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Generally, if X is a discrete random variable, and the probability distribution of X is

X	x_1	x_2	$\cdots$	x_n	$\cdots$
P_k	p_1	p_2	$\cdots$	p_n	$\cdots$

Then, the probability distribution of $Y = g(X)$ is:

Y	$g(x_1)$	$g(x_2)$	$\cdots$	$g(x_n)$	$\cdots$
P_k	p_1	p_2	$\cdots$	p_n	$\cdots$

Attention: If some of $g(x_k)$ are the same, they need to be appropriately combined.

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Example let r.v.X pdf be like

$$f_X(\boldsymbol{x})=\begin{cases}\dfrac{x}{8}, & 0<x<4,\\ 0, & else\end{cases}$$

then find the pdf of r.v.$\boldsymbol{Y}$,which $\boldsymbol{Y}=\mathbf{2}\boldsymbol{X}+\mathbf{8}$.

Solution: **(1) find the cdf of Y**

(2) find the pdf of Y

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Solution: (1) find the cdf of Y

$$F_Y(y)=P(Y\leqslant y)=P(2X+8\leqslant y)=P(X\leq\frac{y-8}{2})=\int_{-\infty}^{\frac{y-8}{2}}f_X(x)dx$$

(2) find the pdf of Y

$$f_Y(y)=F'_Y(y)=[\int_{-\infty}^{\frac{y-8}{2}}f_X(x)dx]'=f_X\left(\frac{y-8}{2}\right)[\frac{y-8}{2}]'$$

Finally,

$$f_Y(y)=\begin{cases}\frac{1}{8}\cdot\frac{y-8}{2}\cdot\frac{1}{2}, & 0<\frac{y-8}{2}<4,\\ 0, & else\end{cases}=\begin{cases}\frac{y-8}{32}, & 8<y<16,\\ 0, & else\end{cases}$$

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Example Suppose r.v. X has a uniform distribution on the interval $(0, 1)$, find the pdf of $Y = -\frac{1}{2}\ln(1 - X)$.

Solution: The probability density function is

$$f_X(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

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Solution: **(1) find the cdf of Y**

$$F_Y(y) = P\{Y \le y\} = P\left\{-\frac{1}{2}\ln(1 - X) \le y\right\}$$

$$= P\left\{X \le 1 - e^{-2y}\right\} = F_X\left(1 - e^{-2y}\right)$$

(2) find the pdf of Y

$$f_Y(y) = \mathrm{f}_X\left(1 - e^{-2y}\right) \times 2e^{-2y}$$

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Then

$$f_Y(y) = \mathrm{f}_X\left(1 - e^{-2y}\right) \times 2e^{-2y}$$

So $f_Y(y) = 2e^{-2y}$, and $0 < 1 - e^{-2y} < 1 \Leftrightarrow y > 0$. Finally we have

$$f_Y(y) = \begin{cases} 2e^{-2y}, & y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

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Example Suppose the pdf of r.v. X is $f_X(x)$. Find the pdf of $Y = X^2$.

Solution: When $y \leq 0$, we have

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = 0$$

When $y > 0$,

$$F_Y(y) = P\{Y \leq y\} = P\{X^2 \leq y\}$$

$$= P\{-\sqrt{y} < X < \sqrt{y}\} = F_X(\sqrt{y}) - F_X(\sqrt{y})$$

Then $f_Y(y) = f_X(\sqrt{y}) \times \frac{1}{2\sqrt{y}} + f_X(-\sqrt{y}) \times \frac{1}{2\sqrt{y}}$
$= \frac{1}{2\sqrt{y}}\left[f_X(\sqrt{y}) + f_X(-\sqrt{y})\right]$
From what has been discussed above, we have

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}}\left[f_X(\sqrt{y}) + f_X(-\sqrt{y})\right], & y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

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(1)Find the cdf of Y=g(X)

① $\Omega_X \rightarrow \Omega_Y$;

② $Y=f(x) \rightarrow G(y)$;

③ $F_Y(y) \rightarrow y \in R$

(2) Find the pdf of Y

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Theorem: The pdf of r.v. X is $f_X(x)$. The function $g(x)$ is monotonic and differentiable, then $g(x)$ has an inverse function $h(y)$, and $h(y)$ is differentiable, then $Y = g(X)$ is a continuous r.v., and its pdf is

$$f_Y(y)=\begin{cases} f_X(h(y))\cdot|h'(y)| & \alpha < y < \beta \\ 0 & \text{others} \end{cases}$$

$$\alpha = \min(g(-\infty), g(+\infty)),$$

$$\beta = \max(g(-\infty), g(+\infty))$$

$h(y)$ is the inverse function of $g(x)$.

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Proof: Let $g(x)' > 0$, then $g(x)$ increases strictly monotonically in $(-\infty, +\infty)$, its inverse function $h(y)$ exists, and increases strictly monotonically in (α, β), which is differentiable.

(1) Find out the distribution function $F_Y(y)$ of $y = g(x)$:

$y = g(x)$ take values in (α, β),

When $y < \alpha$, $F_Y(y) = P(Y \le y) = 0$

When $y \ge \beta$, $F_Y(y) = P(Y \le y) = 1$

When $\alpha \le y < \beta$, $F_Y(y) = P(Y \le y)$

$$= P(g(X) \le y)$$

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Inverse function

Distribution function definition

$$= P[X \le h(y)] = \int_{-\infty}^{h(y)} f_X(x)dx = F_X[h(y)]$$

(2) Find out the probability density of $y = g(x)$:

The probability density of Y can be obtained by deriving $F_Y(y)$ with y:

$$f_Y(y) = \begin{cases} f_X[h(y)] \cdot h'(y) & \alpha < y < \beta \\ 0 & \text{others} \end{cases}$$

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(3) When $g(x)' < 0$, similarly,

$$f_Y(y) = \begin{cases} f_X[h(y)][-h'(y)] & \alpha < y < \beta \\ 0 & \text{others} \end{cases}$$

(4) Above, the pdf of $Y = g(X)$ is:

$$f_Y(y) = \begin{cases} f_X[h(y)] \cdot |h'(y)| & \alpha < y < \beta \\ 0 & \text{others} \end{cases}$$

$$\alpha = \min(g(a), g(b)),\ \beta = \max(g(a), g(b))$$

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Example Suppose r.v. X obey a uniform distribution on (0,1). Find the pdf of $Y = -2\ln X$.

Solution: $X \in (0, 1)$, $\ln x < 0$

$$y = -2\ln x > 0 \Rightarrow \quad y' = -\frac{2}{x} < 0$$

Then y decreases monotonically on (0,1) and has an inverse function:

$$x = h(y) = e^{-y/2}$$

Bring in formula, then:

$$f_Y(y) = \begin{cases} f_X(e^{-y/2})\left|\dfrac{d(e^{-y/2})}{dy}\right| & 0 < e^{-y/2} < 1 \\ 0 & \text{others} \end{cases}$$

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$$f_Y(y)=\begin{cases} f_X(e^{-y/2})\left|\dfrac{d(e^{-y/2})}{dy}\right|, & 0<e^{-y/2}<1 \\ 0, & \text{others} \end{cases}$$

X obey a uniform distribution on (0,1)， so,

$$f_X(x)=\begin{cases} 1, & 0<x<1 \\ 0, & \text{others} \end{cases}$$

Thus ,
$$f_Y(y)=\begin{cases} \dfrac{1}{2}e^{-y/2} & y>0 \\ 0 & \text{others} \end{cases}$$

Y obeys an exponential distribution with a parameter of 1/2.

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Example $X \sim N(\mu,\sigma^2)$, **Find pdf of** $Y = a + bX$.

Solution:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\Rightarrow \quad x = h(y) = \frac{y-a}{b} \qquad h'(y) = \frac{1}{b}$$

$$\Rightarrow \quad f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma} \cdot \left|\frac{1}{b}\right| \cdot e^{-\left(\frac{y-a}{b}-\mu\right)^2 \Big/ 2\sigma^2}$$

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$$f_Y(y)=\frac{1}{\sqrt{2\pi}\sigma}\cdot\left|\frac{1}{b}\right|\cdot e^{-\left(\frac{y-a}{b}-\mu\right)^2\Big/2\sigma^2}$$

$$=\frac{1}{|b|}\cdot\frac{1}{\sqrt{2\pi}\sigma}e^{-[y-(a+b\mu)]^2\Big/2\sigma^2b^2}$$

$$=\frac{1}{\sqrt{2\pi}\underline{\sigma|b|}}\cdot e^{-[y-(\underline{a+b\mu})]^2\Big/2\underline{\sigma^2b^2}}$$

$$\boxed{Y=a+bX\sim N(a+b\mu,\ \sigma^2b^2),}$$

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$$X \sim N(\mu, \sigma^2)$$

$$Y = a + bX \sim N(a + b\mu,\ \sigma^2 b^2),$$

The linear function of normal distribution still obeys normal distribution. Especially, when

$$b = \frac{1}{\sigma},\ a = -\frac{\mu}{\sigma},\quad Y = \frac{X - \mu}{\sigma} \sim N(0,1)$$

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Examples

Example Suppose pdf of r.v.X is:

Find: pdf of $Y = \sin X$.

$$f(x) = \begin{cases} \dfrac{2x}{\pi^2} & 0 < x < \pi \\ 0 & \text{others} \end{cases}$$

Solution: $0 < x < \pi \qquad 0 < y \le 1$

When $y < 0$, $F_Y(y) = 0$;

When $y \ge 1$, $F_Y(y) = 1$;

When $0 \le y < 1$, $F_Y(y) = P(Y \le y) = P(\sin X \le y)$

$$= P(0 \le X \le \arcsin y) + P(\pi - \arcsin y \le X \le \pi)$$

$$= \int_0^{\arcsin y} \frac{2x}{\pi^2} dx + \int_{\pi - \arcsin y}^{\pi} \frac{2x}{\pi^2} dx$$

$$= \left(\frac{\arcsin y}{\pi}\right)^2 + 1 - \left(\frac{\pi - \arcsin y}{\pi}\right)^2$$

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Examples

$$f_Y(y) = \frac{dF_Y(y)}{dy}$$

Thus, the pdf of $Y = \sin X$ obtained by derived from $F_Y(y)$ is:

$$f_Y(y) = \begin{cases} \dfrac{2}{\pi\sqrt{1-y^2}} & 0 < y < 1 \\ 0 & \text{others} \end{cases}$$